

Homework 5- Due Date April 13, 11:59 P.M.

Problem 1. Sarah works as a sales representative selling house insurance over the phone. Only 10% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.

- (a) What is the probability that Sarah makes her first sale on an odd-numbered call?
- (b) What is the probability that Sarah makes her second sale on an even-numbered call?

Problem 2. Let X and Y be two independent random variables, X has a Geometric distribution with the probability of “success” $p = 1/3$, Y has a Poisson distribution with mean 3. Find $P(X = Y)$

Problem 3. In this problem you may neglect the probability of twins.

- (a) A family has 5 children. Find the probability that they have 2 boys and 3 girls.
- (b) A family decides to have children until they have 3 girls. Find the probability that they have 2 boys.
- (c) A family decides to have children until they have 3 girls. Let C be the total number of children in the family. Compute $E[C]$ and $V(C)$.

Problem 4. Consider a random variable X with $E[X] = 4$ and $\text{Var}[X] = 9$.

- (a) Calculate $P(|X - 1| < 3)$ if X follows a normal distribution. Write your final answer based on the CDF of standard normal.
- (b) Find $P(1 < X^2 < 4)$. Write your final answer based on the CDF of standard normal.

Problem 5. Between the hours of 2 and 4p.m. the average number of phone calls per minute coming into the switch-board of a company is 2.5. Find the probability that during one particular minute there will be

- (a) 0,
- (b) 1,

- (c) 2,
- (d) 3 or fewer,
- (f) more than 3 phone calls.

Problem 6. The number of typos made by Sarah follows Poisson distribution with the rate of 0.4 typos per page. Assume that the numbers of typos on different pages are independent.

- (a) Find the probability that there are exactly 4 typos in a 10-page lecture.
- (b) Find the probability that there is at least one page with no typos in a 10-page lecture.
- (c) Find the probability that there are exactly three pages with no typos in a 10-page lecture.